Qubit Reset Channel

$$\Lambda(\rho) = Tr(\rho)\,|0\rangle\langle 0|$$

$$(A, B) \to |\phi^+\rangle\langle\phi^+|$$

$$(\Lambda(A), B) \to \Lambda(A) \otimes \underbrace{\rho_B}_{Tr_A(\rho)} = |0\rangle\langle 0| \otimes \frac{I}{2}$$

Completely Dephasing Channel

$$\Delta\begin{pmatrix}\alpha_{00} & \alpha_{01}\\ \alpha_{10} & \alpha_{11}\end{pmatrix} = \begin{pmatrix}\alpha_{00} & 0\\ 0 & \alpha_{11}\end{pmatrix}$$

$$(A,B) \to |\phi^+\rangle\langle\phi^+| = \begin{pmatrix}1/2 & 0 & 0 & 1/2\\ 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\\ 1/2 & 0 & 0 & 1/2\end{pmatrix} = \sum_{a,b\in\{0,1\}} \frac{1}{2}|a\rangle\langle b| \otimes |a\rangle\langle b|$$

$$(\Delta(A),B) \to \frac{1}{2}|0\rangle\langle 0| \otimes |0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| \otimes |1\rangle\langle 1|$$

$$= \begin{bmatrix}\Delta\begin{pmatrix}1/2 & 0\\ 0 & 0\end{pmatrix} & \Delta\begin{pmatrix}0 & 1/2\\ 0 & 0\end{pmatrix}\\ \Delta\begin{pmatrix}0 & 0\\ 1/2 & 0\end{pmatrix} & \Delta\begin{pmatrix}0 & 0\\ 0 & 1/2\end{pmatrix}\end{bmatrix}$$

$$\Delta_\varepsilon = (1-\varepsilon)\,\mathrm{Id} + \varepsilon\Delta \qquad \Delta_\varepsilon\begin{pmatrix}\alpha_{00} & \alpha_{01}\\ \alpha_{10} & \alpha_{11}\end{pmatrix} = \begin{pmatrix}\alpha_{00} & (1-\varepsilon)\alpha_{01}\\ (1-\varepsilon)\alpha_{10} & \alpha_{11}\end{pmatrix}$$

Compeletely Depolarising Channel

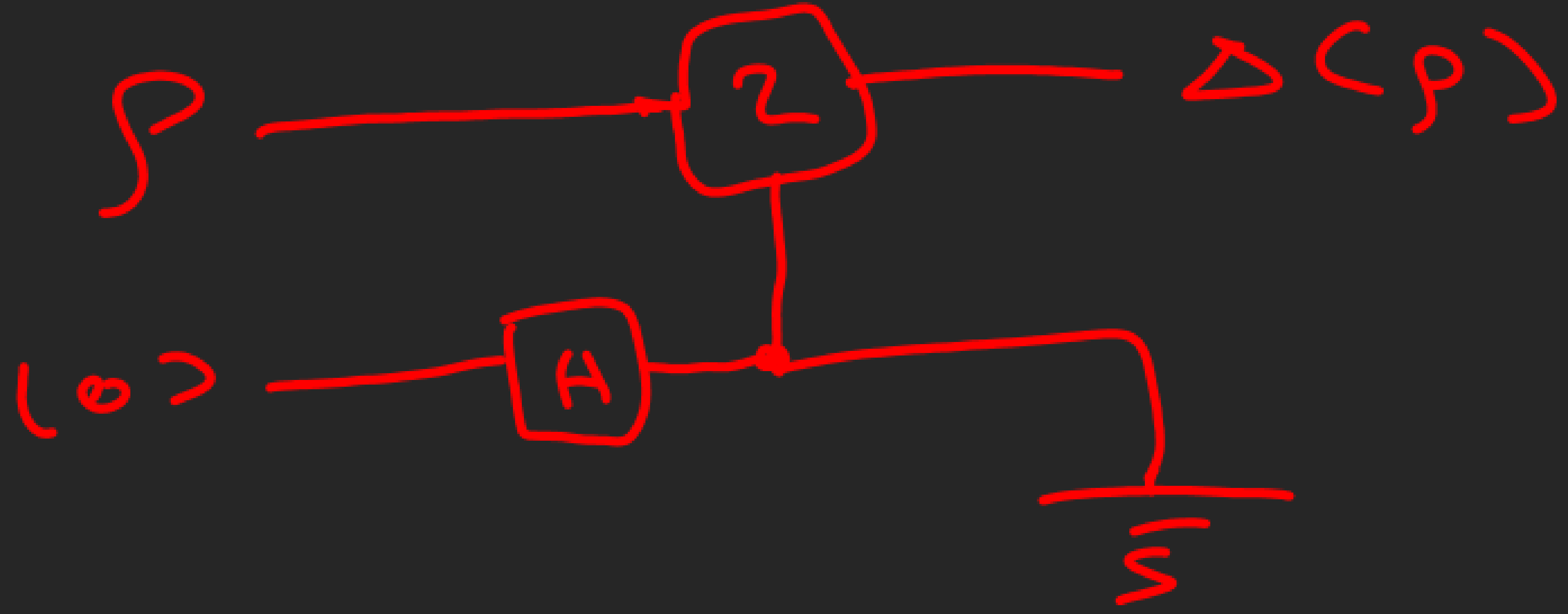
$$\Omega(\rho) = Tr(\rho) \cdot \frac{I}{2}$$

$$\Omega_\varepsilon = (1-\varepsilon) \cdot Id + \varepsilon \, \Omega$$

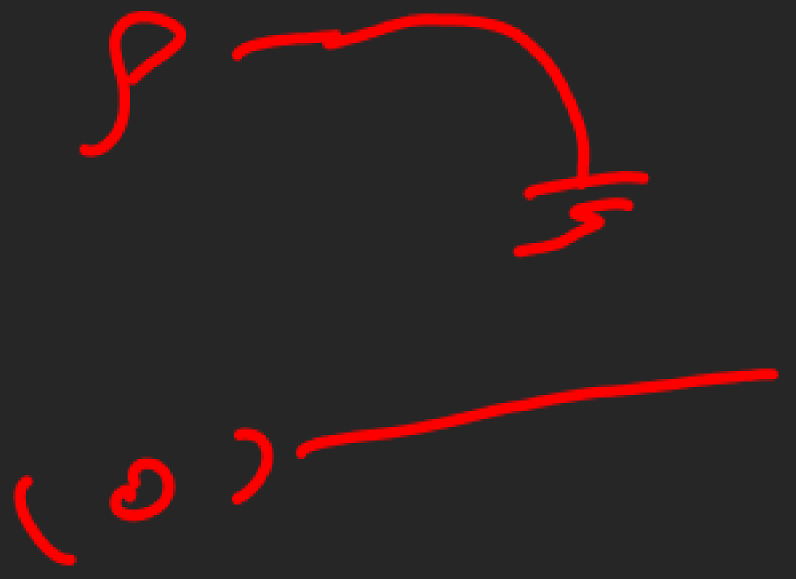
Representations

Stinespring

OR



OR



$\Lambda(\rho) = \mathrm{Tr}(\rho)|0\rangle\langle 0|$

Kraus

$$\left.\begin{array}{l} A_0 = |0\rangle\langle 0| \\ A_1 = |1\rangle\langle 1| \\ \text{OR} \\ A_0 = \frac{I}{\sqrt{2}} \\ A_1 = \frac{\sigma_z}{\sqrt{2}} \end{array}\right\} \to \Delta(\rho)$$

$$\left.\begin{array}{l} A_0 = |0\rangle\langle 0| \\ A_1 = |0\rangle\langle 1| \end{array}\right\} \to \Lambda(\rho)$$

$$\left.\begin{array}{l} A_0 = \frac{|0\rangle\langle 0|}{\sqrt{2}} \quad A_1 = \frac{|0\rangle\langle 1|}{\sqrt{2}} \\ A_2 = \frac{|1\rangle\langle 0|}{\sqrt{2}} \quad A_3 = \frac{|1\rangle\langle 1|}{\sqrt{2}} \\ \text{OR} \\ A_0 = \frac{I}{2} \quad A_1 = \frac{\sigma_x}{2} \\ A_2 = \frac{\sigma_y}{2} \quad A_3 = \frac{\sigma_z}{2} \end{array}\right\} \to \Omega(\rho)$$

$A_0 = U \to$ unitary Channels

Choi

$$J(\Delta) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$J(\Omega) = \begin{pmatrix} 1/2 & 0 & 0 & 0 \\ 0 & 1/2 & 0 & 0 \\ 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 1/2 \end{pmatrix}$$

$$J(\Lambda) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$J(Id) = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}$$